

Illustration of Advanced Prompts Utilized by LLM Inference on the Cora Dataset

Direct: Given the information of the node: `<raw_text>`. Question: Which of the following categories does this paper belong to? Here are the categories: `<labels>`. Reply only with one category that you think this paper might belong to. Only reply with the category name without any other words.

Chain-of-Thought: Given the information of the node: `<raw_text>`. Question: Which of the following types does this paper belong to? Here are the 7 categories: `<labels>`. **Let's think about it step by step.** Analyze the content of the node and choose one appropriate category. Output format: `<reason: >`, `<classification: >`

Tree-of-Thought: Given the information of the node: `<raw_text>`. Imagine three different experts answering this question. All experts will write down 1 step of their thinking, and then share it with the group. Then all experts will go on to the next step, etc. If any expert realizes they're wrong at any point then they leave. Question: Based on this information, which of the following categories does this paper belong to? Here are 7 categories: `<labels>`. **Let's think through this using a tree of thought approach.** Output format: `<discussion: >`, `<classification: >`. The classification should only consist of one of the category names listed.

ReACT: Given the information of the node: `<raw_text>`. Your task is to determine which of the following categories this paper belongs to. Here are the 7 categories: `<labels>`. **Solve this question by interleaving the Thought, Action, and Observation steps.** Thought can reason about the current situation, and Action can be one of the following: (1) Search[entity], which searches the exact entity on Wikipedia and returns the first paragraph if it exists. If not, it will return some similar entities to search. (2) Lookup[keyword], which returns the next sentence containing the keyword in the current passage. (3) Finish[answer], which returns the answer and finishes the task. The output format must be `<process: >`, `<classification: >`. The classification should only consist of one of the category names listed.